



Course: Chemical Evolution of Galaxies

Is metallicity a reliable indicator of the age and star formation history of a galaxy?

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Understanding the evolution of galaxies involves deciphering multiple properties. One key parameter often examined in this context is metallicity. This property is often related to other equally important such as; the age and the star formation history (SFH). The chemical enrichment of galaxies over time reflects the cumulative history of star formation and supernova events, with younger stars generally exhibiting higher metallicities due to the progressive build-up of heavy elements. This debate is about whether the metallicity serves as a good estimator of age and star formation history for galaxies.

Metallicity serves as a crucial, though not infallible, indicator of both the age and SFH of a galaxy. The age-metallicity relationship (AMR) has been employed to study the chemical evolution of different galactic and extragalactic stellar systems (Catelan, de Freitas Pacheco, 1994). Furthermore, the variation in metallicity across a galaxy and its correlation with different star formation episodes provide insights into the galaxy's SFH, revealing periods of intense star formation, quiescence, and the influence of external processes such as gas inflows and outflows. The fundamental metallicity relation (FMR) has shown that in the local universe a relation exists between metallicity, stellar mass and SFR (Mannucci et al., 2010).

With respect to the metallicity and age of a galaxy. There is a trend for the younger galaxies to have more metal-rich stellar populations. This is as would be expected if the more recent starbursts were occurring in progressively more enriched gas (Terlevich, Forbes, 2002).

Relations between metallicity, age and stellar mass hold for both early- and late-type galaxies. However, the age range of the sample is less mass-dependent for early-type than for late-type galaxies. At masses below $10^{10} M_{\odot}$, late-type galaxies are younger and more metal-poor than early-type galaxies. At masses above $10^{11} M_{\odot}$ the metallicities and ages of late-type galaxies are similar to those of early-type galaxies. The above results indicate that young, metal-poor stellar populations are found predominantly in low-mass galaxies. This is consistent with a downsizing scenario, in which the mass and luminosity of the galaxies undergoing active star formation become progressively lower as the universe becomes older (Cowie et al., 1996). Initially, this would imply that the low metallicities of low-mass galaxies just reflect the fact that these galaxies have had less time to produce metals (Gallazzi et al., 2005).

Alternatively, metallicity has been shown to be particularly sensitive to their SFH (Sommariva, V. et al., 2012). The observational sample exhibited in Yates et al. (2012), showed a reversal in the metallicity-SFR dependence from lower to higher stellar masses. At low masses, low-SFR galaxies show higher metallicities with respect to star forming galaxies. Higher masses show lower metallicities. Interpreting these results requires considering several factors. Although metallicity is primarily influenced by star formation, infall, and outflow, predicting its dependence on SFR is challenging due to the complex interplay of these processes and their varying impacts on galactic properties (Mannucci et al., 2010). The dependence can be explained by considering metal-rich outflows that have mass-dependence. We find that, considering that outflows are more effective at lower masses ($\sim 10^9 M_{\odot}$), low-SFR galaxies produce fewer SNe, which disturbs less the ISM and consequently expelling metals less efficiently (e.g. Mac Low, Ferrara (1999)). At higher masses ($\sim 10^{10.2} M_{\odot}$), outflows become increasingly weak and low-SFR produce less metals and end up impoverishing the ISM with respect to star forming galaxies. However the cosmological semi-analytic model of Yates et al. (2012) shows a decrease in metallicity with respect to stellar mass considering a fixed SFR over $\sim 10^{10.4} M_{\odot}$. This cannot be explained only by mass-dependant outflows and it might be explained by metal-poor galactic infall at high masses.

There are, however, factors hampering in Yates et al. (2012) interpretation. The main one being the dependence of metallicity on SFR is itself strongly dependent on how these properties are measured and although a high-mass dependence is undeniable in the simulated sample, it is not present in the observational sample. Moreover, the dependence of outflows on galactic properties is not well understood. In many models, star formation triggers supernovae, which inject energy, radiation, and momentum into the interstellar medium, expelling some of the enriched gas (Spitoni et al., 2010). Also, infalls are expected to influence metallicity in two ways. The infall of metal-poor gas can dilute metal-rich gas, reducing observed metallicity (see Finlator, Davé (2008)) and it can trigger star formation (Schmidt, 1959), with new stars enriching

the gas with metals and increasing metallicity. If galaxy mergers, rather than smooth infall from the intergalactic medium, drive gas into galaxies, the inflowing gas can be significantly enriched, potentially negating the dilution effect. Some semi-analytical models of galaxy formation (de Rossi et al., 2007) suggest that metallicity is mainly influenced by star formation efficiency dependent on galaxy mass, with less massive galaxies being less evolved and exhibiting lower metallicities.

Eventhough metallicity can be used as an indicator of age and SFH of galaxies, the complexity of metallicity gradients affect the reliability of it. The presence of multiple stellar populations, and environmental effects necessitate a multi-faceted approach. Therefore, while metallicity is a significant and insightful indicator, it is most effective when integrated with a broader array of observational data, ensuring a nuanced and comprehensive understanding of a galaxy's age and star formation history. Combining metallicity data with complementary observational methods, such as stellar population studies, spectroscopic analysis, and star formation rate measurements, enhances the reliability of reconstructing a galaxy's evolutionary history.

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